

CT ADEQUACY ANALYSIS REPORT

Device: RED670

Report Ref: CTA-20260729

Date Issued: July 29, 2026

Revision	Prepared By	Checked By	Approved By
Rev. 0	To Be Confirmed	To Be Confirmed	To Be Confirmed

1.0 PROJECT & CLIENT INFORMATION

Client Name: Confidential Client	Project / Site Company: Project Site Company Name Not Specified
Project Start Date: To Be Confirmed	Expected Completion Date: To Be Confirmed

2.0 EXECUTIVE SUMMARY

This report presents the current transformer (CT) adequacy assessment for "RED670" (Protection Function: RED670). The purpose of this check is to confirm whether the installed/selected CT can deliver sufficient secondary voltage (Vk) / accuracy limit factor to drive connected protection relays correctly under system fault conditions, without core saturation. Based on the inputs and calculations detailed in this report, the CT meets the required knee-point voltage (Vk) / accuracy limit factor and is suitably dimensioned for the fault conditions evaluated. All input data, formulas applied, intermediate values and final results are presented as actually computed for this device, in full, for independent verification by engineering and review teams.

SUITABLY DIMENSIONED

Vk Required: 400.05 V | Vk Available: 1250 V

CT Adequacy Analysis Report (continued)

3.0 INPUT PARAMETERS (AS ENTERED)

Parameter	Value	Unit	Description
CT Ratio	1800/1	A	Current transformer turns ratio (primary/secondary ampere rating). Determines the scaling factor between primary and secondary circuits.
Accuracy Class	PX		CT accuracy classification per IEC or IEEE standards, indicating the maximum permissible error under specified conditions.
CT Resistance	5.6	ohm	Secondary winding resistance of the current transformer, affecting the voltage drop across the CT itself.
Lead Resistance	1.1	ohm	Total loop resistance of the pilot wire/cable connecting the CT secondary to the relay, as computed from the route length and cable data provided.
Relay Burden	0.02	VA	Apparent power consumption of the connected relay or meter at rated secondary current, expressed in volt-amperes.
Bus Voltage	132	kV	System operating voltage at the point where the CT is installed, used for fault level calculations.
Fault Level	50	kA	Maximum fault current that can occur at the CT location, determining the worst-case secondary current for adequacy assessment.
Route Length	1.74	km	Physical distance from CT to relay location, used to compute lead resistance.

4.0 CALCULATION BREAKDOWN — RED670

Fault Condition	Formula Applied (as computed)	Ealreq (V)	Vk Req (V)
Differential Close-in	$Ealreq = 1800 \times (1/1800) \times 6.72 = 186.67$	186.67	149.34
Differential Through 3-ph	$Ealreq = 2 \times 3037.49 \times (1/1800) \times 6.72 = 315.32$	315.32	252.26
Differential Through 1-ph	$Ealreq = 2 \times 3126.96 \times (1/1800) \times 6.72 = 324.61$	324.61	259.69
Distance Close-in	$Ealreq = 1800 \times (1/1800) \times 1 \times 6.72 = 186.67$	186.67	149.34
Distance Endzone-1 3-ph	$Ealreq = 4700 \times (1/1800) \times 3 \times 6.72 = 488.15$	488.15	390.52
Distance Endzone-1 1-ph (MAX)	$Ealreq = 4815 \times (1/1800) \times 3 \times 6.72 = 500.06$	500.06	400.05

FINAL RESULT: Ealreq Max = 500.06 V Vk Required = 400.05 V

5.0 NOTES & ASSUMPTIONS

All resistance and current values shown are as computed by the calculation engine from the inputs provided; none are default or placeholder values.

Lead resistance is computed from the route length and cable data supplied for this device/circuit.

Vk Available reflects the manufacturer-declared or test-certificate value supplied for this device; where unavailable, the verdict is reported as NOT APPLICABLE.

This assessment covers CT knee-point voltage / Kssc adequacy only and does not constitute a complete protection coordination study.

CT Adequacy Analysis Report (continued)

6.0 FORMULAS & CALCULATION METHOD (VK METHOD)

CT adequacy is verified using the knee-point voltage (Vk) method per IEC 61869-2 and IEEE C57.13 standards:

1. Secondary Burden & Lead Resistance:

$$2RL = 2 \times R_{20} \times (1 + \pm(t - 20)) \times l, \quad Rl = Sl / In^2$$

Where: R_{20} = resistance at 20°C, $\pm = 0.00393/K$, $t = 75^\circ\text{C}$ operating temp, l = lead length in km

2. Differential Protection Ealreq Calculation:

$$\text{Close-in: } E_{alreq} = I_{kmax} \times (I_{sn} / I_{pn}) \times (R_{ct} + Rl + Sr/Ir^2)$$

$$\text{Through (3ph \& 1ph): } E_{alreq} = 2 \times I_{tmax} \times (I_{sn} / I_{pn}) \times (R_{ct} + Rl + Sr/Ir^2)$$

3. Distance Protection Ealreq Calculation:

$$\text{Close-in (a=1): } E_{alreq} = I_{kmax} \times (I_{sn} / I_{pn}) \times a \times (R_{ct} + Rl + Sr/Ir^2)$$

$$\text{Endzone-1 (k=3): } E_{alreq} = I_{kzone1} \times (I_{sn} / I_{pn}) \times k \times (R_{ct} + Rl + Sr/Ir^2)$$

4. Required Knee Point Voltage & Verdict Criteria:

$$V_{k(req)} = E_{alreq(max)} \times 0.8$$

' SUITABLE: Available V_k "e Required V_k

' UNDER-DIMENSIONED: Available $V_k <$ Required V_k

7.0 APPLICABLE STANDARDS & REFERENCES

IEC 61869-2 Instrument transformers: Additional requirements for current transformers.

IS 2705 (Part 1 to 4) Current transformers: General/specific requirements.

IEEE C37.110 Guide for application of current transformers used for protective relaying purposes.

IEEE C57.13 Standard requirements for instrument transformers.

Internal engineering practice for CT knee-point voltage (Vk) adequacy verification.

8.0 REVIEW & APPROVAL

To Be Confirmed

PREPARED BY

Date: _____

To Be Confirmed

CHECKED BY

Date: _____

To Be Confirmed

APPROVED BY

Date: _____